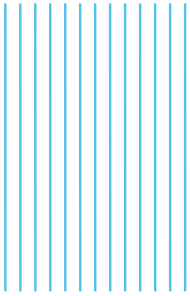


IBAB · TEAM DENOVO

early-stage-bia.onrender.com



BIET

Budget Impact Estimation Tool

Early-stage budget impact analysis for pharmaceutical decision support

THE PROBLEM

Early-stage drug teams cannot answer a payer's first question — what will this cost my budget? — without weeks of bespoke spreadsheet work.

HEADLINE OUTCOME

A five-year model in under 10 minutes instead of weeks, every figure traceable to the input behind it.



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The question every payer asks first

Before a price is agreed, a health insurer wants one number: what does saying yes to this drug do to my budget over the next three to five years?

Why it is hard today

- The model is built from scratch in Excel for every product and market
- Formulas sit in cells no reviewer can trace, so the answer is trusted or rejected whole
- Changing one assumption means rebuilding, so alternatives go untested

Who is affected

- **HEOR & market access**
Must defend every assumption to a payer, with no audit trail to point at
- **Payers & insurers**
Receive a number they cannot interrogate before committing budget
- **Patients**
Access is delayed while the analysis is rebuilt for each market

A budget impact analysis answers affordability — what it costs the budget. That is a different question from cost-effectiveness, which asks whether it is good value. BIET does the first, deliberately.

What is original here

Not a new economic method — a new way of working with one, so the analysis is fast enough to explore and transparent enough to defend.

01

Instant, not re-run

The whole model recomputes in the browser as you type. Zero API calls in the calculation path, so sensitivity is something you feel rather than a job you queue.

02

Evidence beside the model

EviTrack searches PubMed, ClinicalTrials.gov, openFDA, WHO GHO and the World Bank inside the same tool, so an assumption and its source live together.

03

Auditable by construction

Cost components reconcile exactly to scenario totals, and a Methodology view traces every headline figure back to the inputs that produced it.

Why this combination is new

Commercial BIA tools are enterprise licences priced out of early-stage work; academic models are spreadsheets with no interface. BIET puts a live model, its evidence and its audit trail in one browser tab that runs on free infrastructure.

One guided flow, population to decision

Six input tabs become a five-year payer forecast with scenarios, sensitivity and an audit trail — in a single sitting.



Validation runs between every edit and the result: a model that cannot produce a meaningful answer is blocked, with the reason and the field named.

Target users

- HEOR and market-access teams in pharma
- Health-economics researchers and students
- Anyone pricing a therapy before a formal dossier

Key scenario

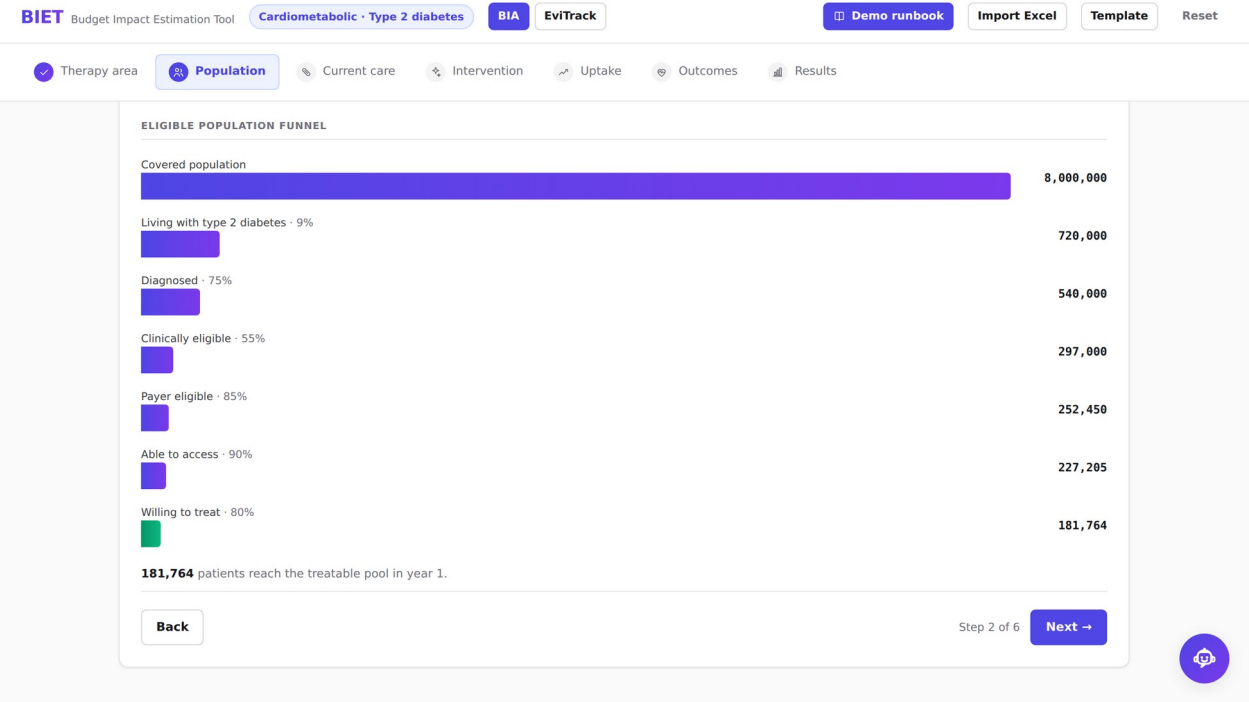
- Pricing a new GLP-1 for a German statutory insurer
- 8 million covered lives, five-year horizon
- Compared against metformin, basal insulin, DPP-4

Expected benefit

- Time — weeks of model building down to minutes
- Quality — validation blocks silent-zero errors
- Outcomes — cost offsets modelled, not omitted

How the number is built

Every currency figure traces back to one funnel: a covered population narrowed step by step to the patients who are actually treated.



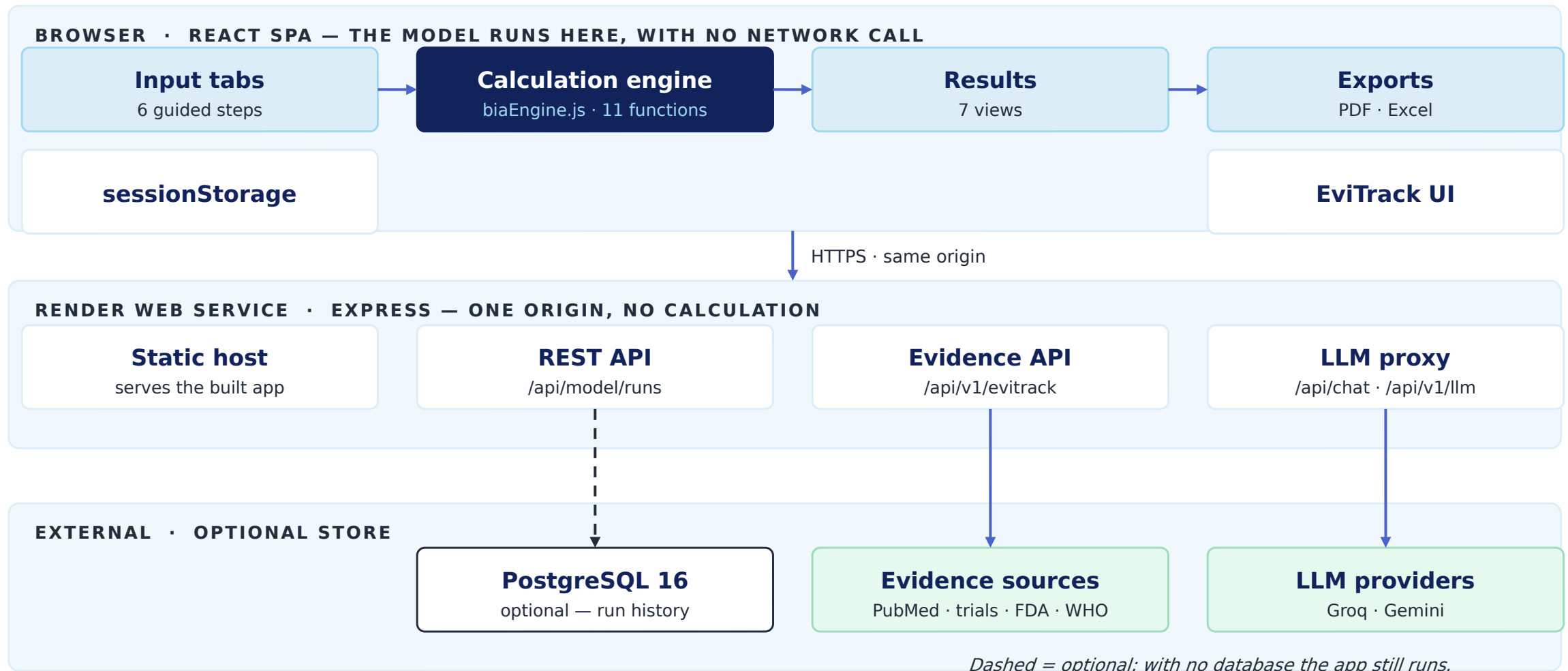
Population tab — each rate is an input; the funnel updates as you type.

Worked example — year 1

Covered population		8,000,000
Living with type 2 diabetes	9%	720,000
Diagnosed	75%	540,000
Clinically eligible	55%	297,000
Payer eligible	85%	252,450
Able to access	90%	227,205
Willing to treat	80%	181,764

A 44-fold reduction. Prevalence and incidence are separate inputs, because a prevalence-only model understates a growing disease.

System architecture





Model, data and how we check it

1 Algorithms & model

- Deterministic cohort projection over the horizon
- Adherence- and persistence-weighted treatment cost
- Relative-risk event offsets, damped by a regain rate
- Binary search for the break-even drug price
- One-way tornado: 10 parameters at $\pm 20\%$
- Undiscounted costs — ISPOR practice for BIA, not CEA

2 Data sources

- PubMed, ClinicalTrials.gov, openFDA — evidence
- WHO GHO, World Bank — epidemiology and expenditure
- Excel import for a team's own parameters
- Every parameter overridable; provenance columns in schema

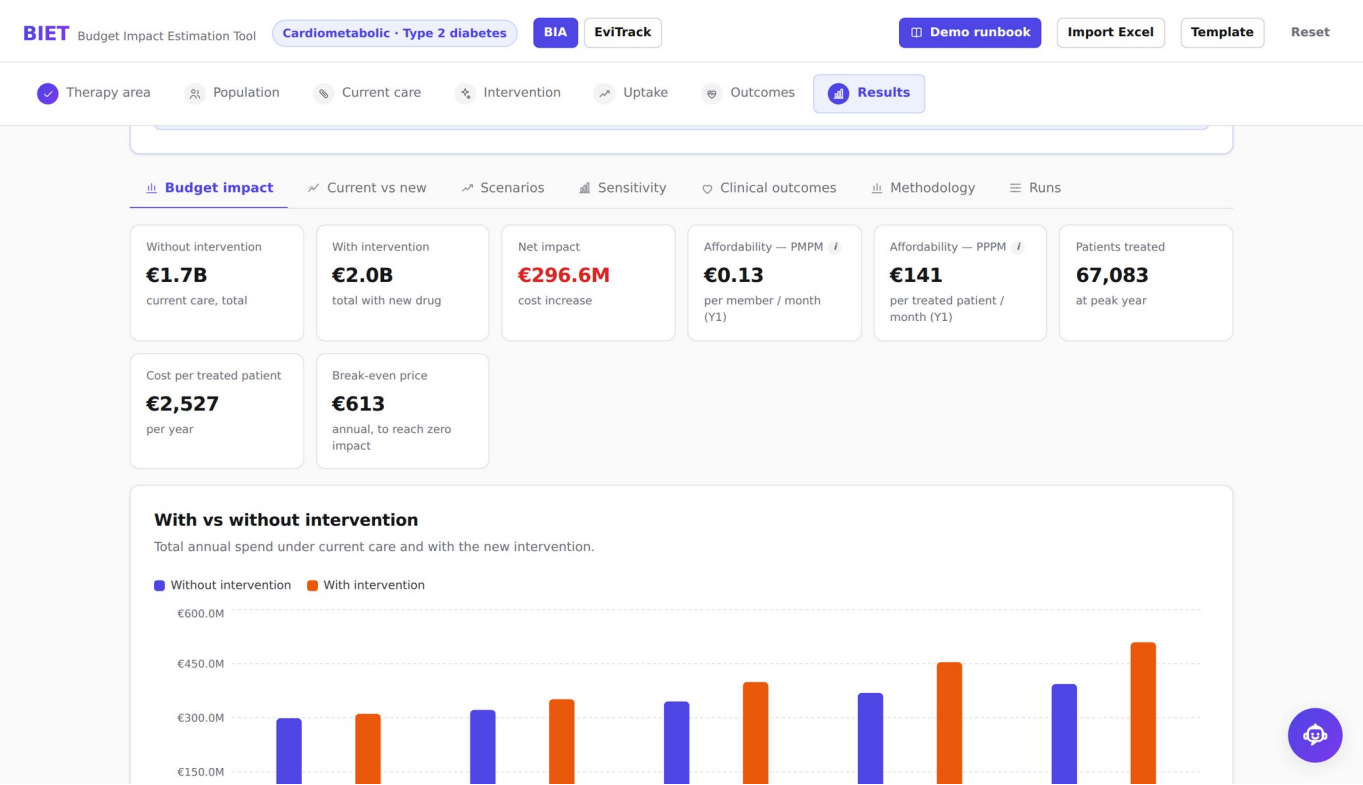
3 Evaluation approach

- 38 automated tests — 14 economics, 24 validation
- Engine pinned to hand-calculable cases, not snapshots
- Cost components reconcile exactly to scenario totals
- Break-even re-solves to a net impact of zero
- 28 invalid inputs probed: 12 blocked, 10 flagged

ALIGNED WITH ISPOR BIA GOOD PRACTICE

Budget-holder perspective · 3-5 year horizon · current vs new treatment mix · undiscounted costs · total and PMPM reporting · scenario analysis

What the tool returns



Live output — Drug X, a GLP-1, in Germany. Demonstration values, not real product data.

€296.6M

net budget impact, 5 years

€0.13

per member per month, year 1

67,083

patients treated at peak

€613

break-even annual price

One walkthrough

- IN**

8,000,000 lives · 9% prevalence · 4→28% uptake · €3,200 drug
- OUT**

181,764 eligible → 67,083 treated → €296.6M = €0.13 PMPM · 2,480 events avoided

Does it hold up when pushed?

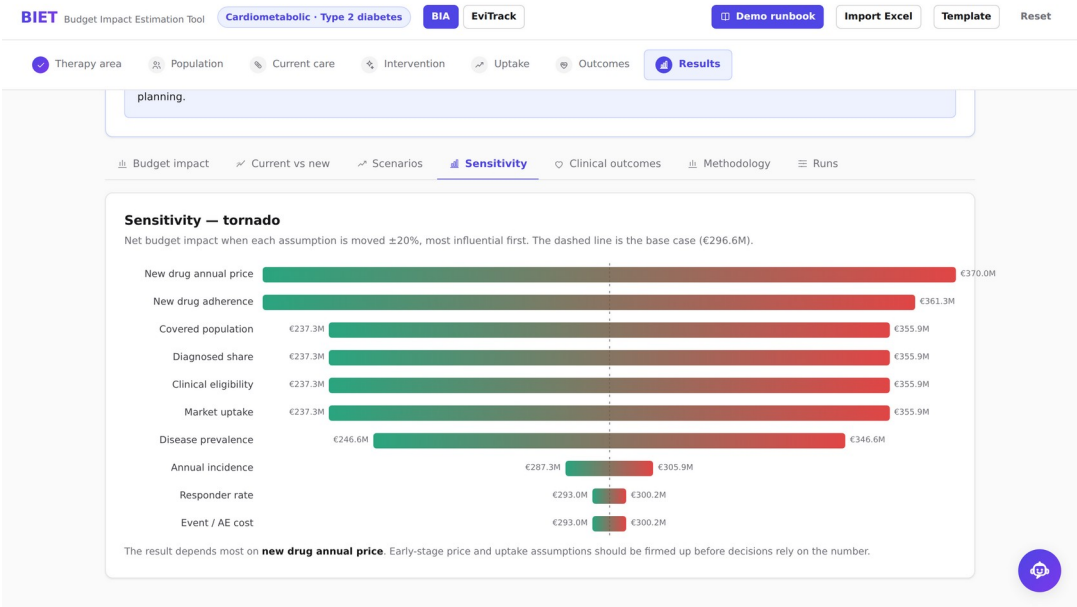
A tool that returns a confident wrong number is worse than one that refuses. We check the arithmetic against a case anyone can verify by hand.

VALIDATION AGAINST A REFERENCE CASE

Covered lives	1,000,000
× prevalence 10%	100,000
× diagnosed 50%	50,000 eligible
× uptake 100%, 1 year	50,000 treated
× new drug ₹100/yr	₹5,000,000
Engine returns	₹5,000,000 ✓

Robustness to changing assumptions

- 28 invalid inputs probed — 12 blocked, 10 flagged, none produce NaN
- Cost components reconcile exactly to scenario totals
- Break-even price re-solves to a net impact of zero



Sensitivity — each parameter moved ±20% alone, sorted by how far the answer swings.

<100ms
recalculation

38
automated tests

0
network calls

Where the assumptions come from

The usual challenge to any model is “where did that number come from?” EviTrack is our answer, inside the same tool.

The screenshot shows the BIET (Budget Impact Estimation Tool) interface. At the top, there's a header with 'BIET Budget Impact Estimation Tool', 'Cardiometabolic · Type 2 diabetes', 'BIA', and 'EviTrack'. Navigation buttons include 'Demo runbook', 'Import Excel', 'Template', and 'Reset'. The main search area has a dropdown for 'AI MODEL' set to 'Not configured' and a search bar containing 'semaglutide' with a 'Search' button. Below the search bar, there are two sections of 'External evidence'. The first section, titled 'External evidence', shows a PubMed result from 2021: 'Safety of Semaglutide.' by Smits MM, Van Raalte DH, with a DOI of 10.3389/fendo.2021.645563. It has a '100% match' and buttons for 'Open source', 'Explain', and '+ Add evidence'. The second section shows a ClinicalTrials.gov result from 2018: 'Semaglutide Effects on Cardiovascular Outcomes (SELECT)', with a '90% match' and similar buttons. Below these is an 'Evidence workspace' section with instructions to 'Curate supporting evidence before using it in your analysis.' and an 'Evidence assistant' section powered by 'GEMINI 3.6 FLASH' that prompts the user to ask questions about the selected evidence.

Five public sources

- **PubMed**
Published literature
- **ClinicalTrials.gov**
Trial registrations & results
- **openFDA**
Label and adverse-event data
- **WHO GHO**
Country epidemiology
- **World Bank**
Population & health expenditure

AI summaries are constrained to the retrieved text and cannot alter any model input.

Current work — what is actually done

Stated honestly, so a reviewer does not have to find the edges themselves.

Working today

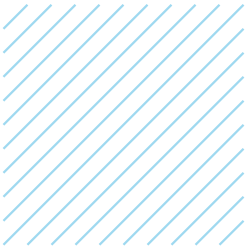
- 3 diseases with subgroups (obesity, T2D, MASH)
- Comparators added or removed as whole columns
- 7 result views incl. tornado and audit trail
- PDF report and 6-sheet Excel export, in-browser
- EviTrack search across 5 public sources
- Run history — compare two models side by side
- Deployed and publicly reachable on Render

Built but not proven

- Parameter provenance schema — 14 tables exist, UI not wired
- Public-data sync — implemented, off by default on free hosting
- AI assistant — works with a key, local answers without one

Known limits & data gaps

- Single country per model — no multi-market roll-up yet
- Subgroups inherit parent parameters rather than their own
- Market share redistributes proportionally, not by displacement
- One-way sensitivity only, no probabilistic run
- Data gap: no validated country cost set — demo values are illustrative
- Data gap: clinical parameters not yet expert-reviewed



Scalability

Three directions, each with the property that already makes it possible.

Computation

Scales because it does not centralise

The model runs on the user's own machine, so a thousand analysts cost the same server capacity as one. The API is only touched for evidence search and persistence.

Disease & market

Scales by data, not by code

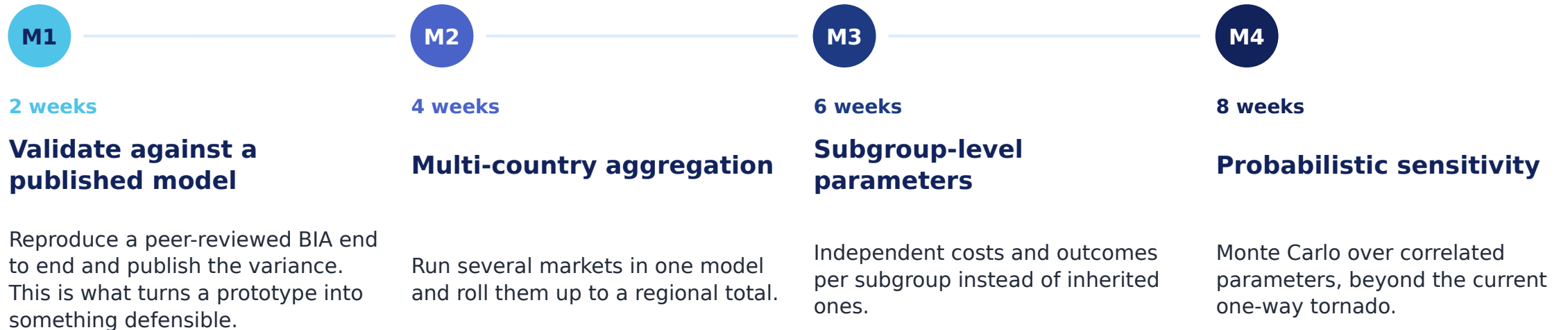
Adding a disease is one registry entry with its subgroups and defaults — no engine change. The same is true of a new comparator or outcome. Multi-country roll-up is the next structural step.

Organisation

Scales because there is nothing to install

It is a browser tab, holds no patient-level data, and degrades gracefully with no database and no API key — so a pilot needs no procurement or infrastructure review.

Future enhancement plan



Dependencies, risks and how we reduce them

Published parameters are paywalled or country-specific

Start from open sources; every parameter is overridable and carries its origin.

Clinical assumptions need expert review

The Methodology view exposes every assumption for an HEOR reviewer to challenge.

Free hosting sleeps; free database expires at 30 days

Already runs with no database at all — search and calculation degrade gracefully.

Compliance if real patient data is ever involved

Cohort-level by design; no patient-level data is stored anywhere.

Who benefits, and how soon

HEOR & market access

A defensible first-pass model in minutes, with sensitivity and an audit trail included rather than added later.

Immediately

Payers & insurers

A number they can interrogate — components reconcile, assumptions are visible, limits are stated.

At pilot

Patients

Faster affordability answers mean less delay between approval and access.

Downstream

Adoption pathway

Pilot inside one therapy-area team on a live pricing question · stakeholders are the HEOR lead, market access and medical affairs · integration effort is low because it is a browser tab with no patient data

€0.13 PMPM

the unit a payer decides on

Minutes, not weeks

live recalculation, zero API calls

Shows its working

reconciles exactly · 38 tests · limits stated